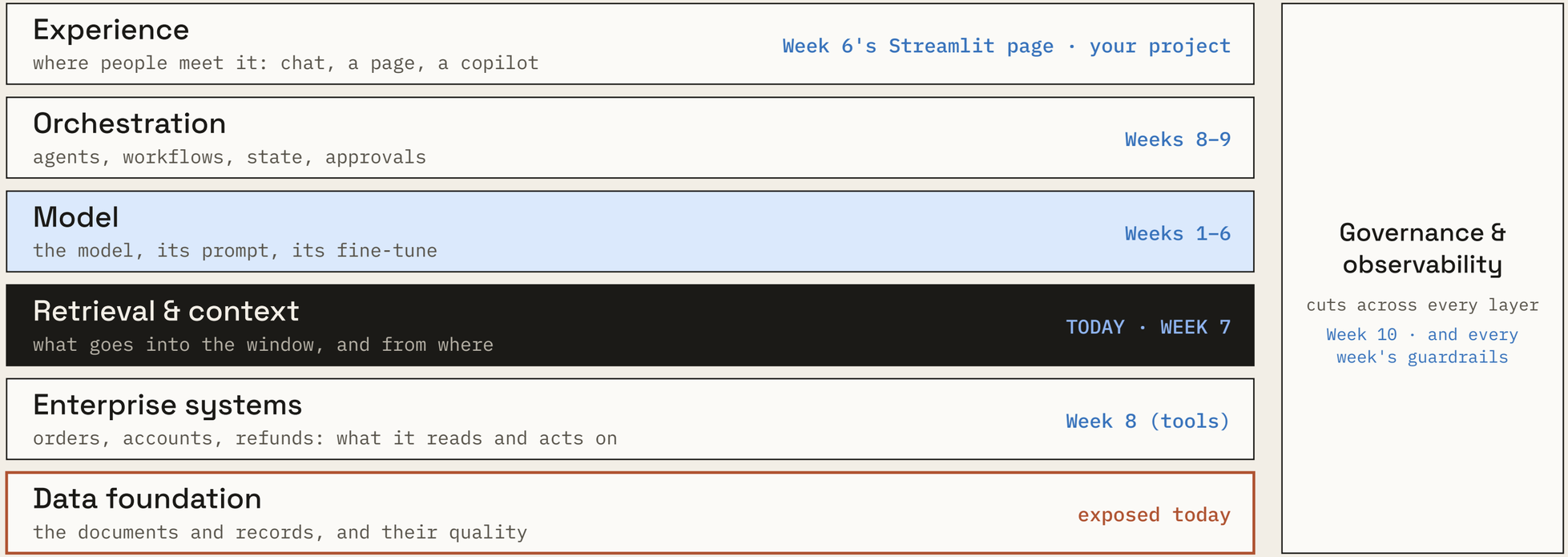
PROMPT AUGMENTATION · RETRIEVAL (RAG) · CONTEXT ENGINEERING

Giving the model the facts

What a model does not know, how to find the right page and put it in the prompt, what goes wrong when the pages are wrong, and how to manage everything the model reads on each call. Measured on one help centre.

An AI system is layers around a model



- A model answers from its **context window**: the text it is given on that call. Today's layer decides what goes in it, and shows how good the documents underneath really are.

Two sources of facts: training and the prompt

IN THE WEIGHTS

Parametric knowledge

learned in training, years ago
a snapshot; changes only by training
Week 6: fine-tune for form, behaviour

IN THE WINDOW

Context knowledge

put in front of it, this call
as fresh as the page you fetched
Week 7: retrieve for facts

WHAT WEEK 6’S MODEL DOES NOT KNOW

It routes NW-40318 perfectly

— and has no idea what Northwind’s return window is, what form LP-14 is for, or where that parcel is.

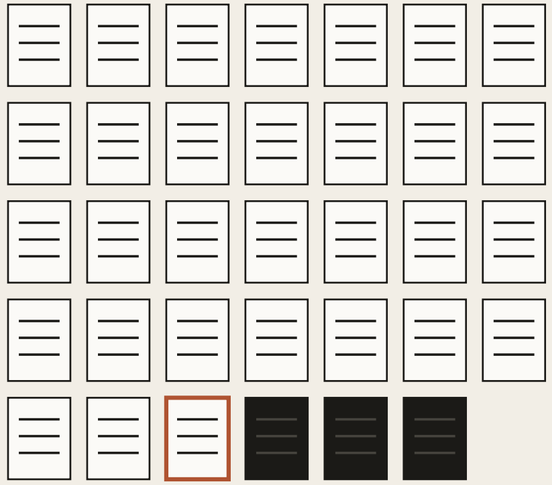
RUNG 0 · train	RUNG 1 · prompt	RUNG 2 · few-shot	RUNG 3 · RETRIEVE	RUNG 4 · fine-tune	RUNG 5 · tools

Week 6’s adaptation ladder: rung 3 is today. It changes the facts put in front of the model, not the model.

- This week the model is **never changed**. Everything new happens before the call: what we put in the prompt.

A help-centre assistant and its test questions

THE HELP CENTRE · 34 ARTICLES



30 public, current

1 old version, never deleted

3 internal, staff only

35 TEST QUESTIONS, ANSWERS KNOWN

8	an exact code or term	
8	said in other words	
5	needs two articles	
3	two parts of one article	
3	an old policy disagrees	
5	only an internal page knows	decline
3	not covered anywhere	decline

27 have an answer in the help centre; 8 must be declined

EVERY DESIGN, SCORED ON

right
of the 27 with an answer

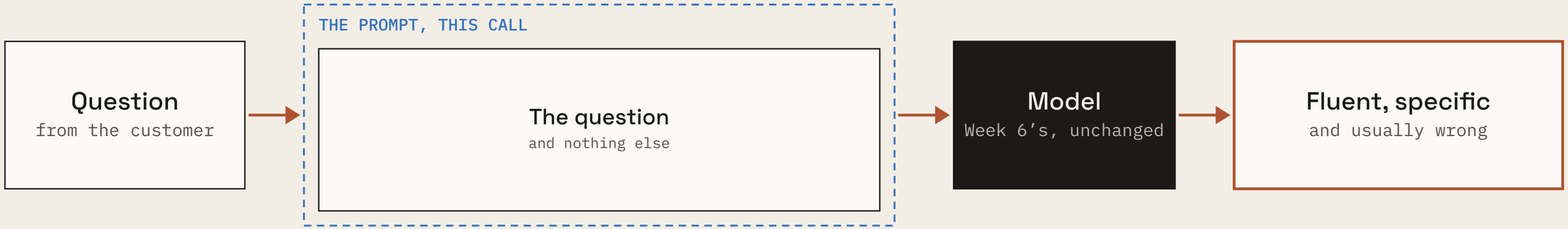
declined
of the 8 with no answer to give

leaked
an internal fact given out

old policy
states the old policy

◆ The same 35 test questions for every design: **27 have an answer** in the help centre and 8 must be declined. ‘Right’ is always counted out of the 27.

Without the facts, the model makes things up



What is form LP-14 for? — MADE UP

Form LP-14 is a lightweight and durable backpack designed specifically for hikers and outdoor enthusiasts who need to carry …

What does error code E-118 mean? — MADE UP

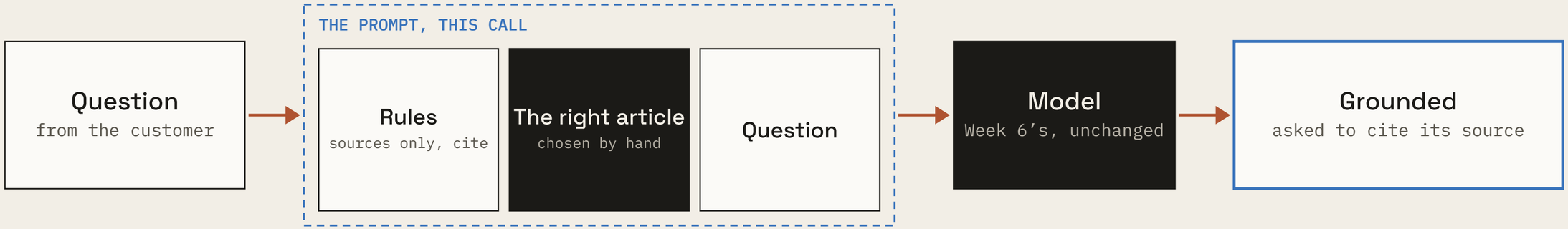
Error code E-118 typically indicates a problem with your computer's network connection or settings. It could be due to …

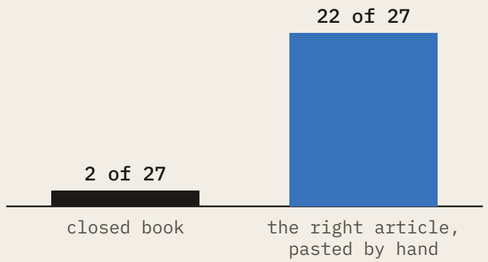
How long do I get to change my mind and send something back? — MADE UP

At Northwind Outfitters, you have 30 days from your purchase date to return items if they don't meet your expectations. This …

- With the fact missing, the likeliest text is a **confident guess**. It does not say 'I don't know'.

Put the right page in the prompt





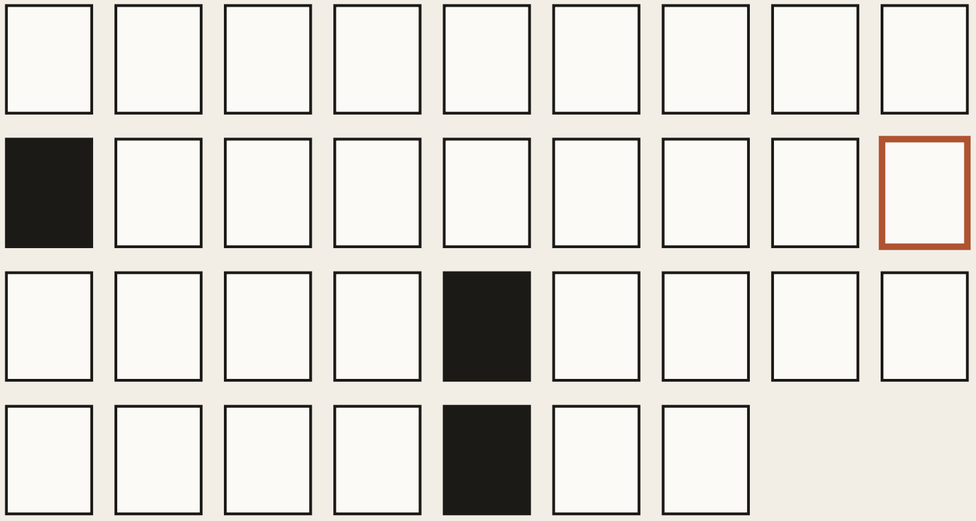
THE BEST CASE, NOT A WORKING SYSTEM

A person chose the right article for every question. Every later design tries to reach this score *without* a person choosing the page.

- The same model went from 2 to **22 of the 27 answerable questions right**, only because the right page was in the prompt.

Why not paste the whole help centre?

THE WINDOW · 10,529 TOKENS, EVERY CALL



+ the question, at the very end

1 · COST AND TIME

10,529 tokens, every call

14.1 s an answer here, against 1.5 s closed book; a rented model bills every token.

2 · IT INCLUDES PAGES IT SHOULD NOT

5 leaks · 1 old-policy answer

The staff pages and the old policy are in the window too.

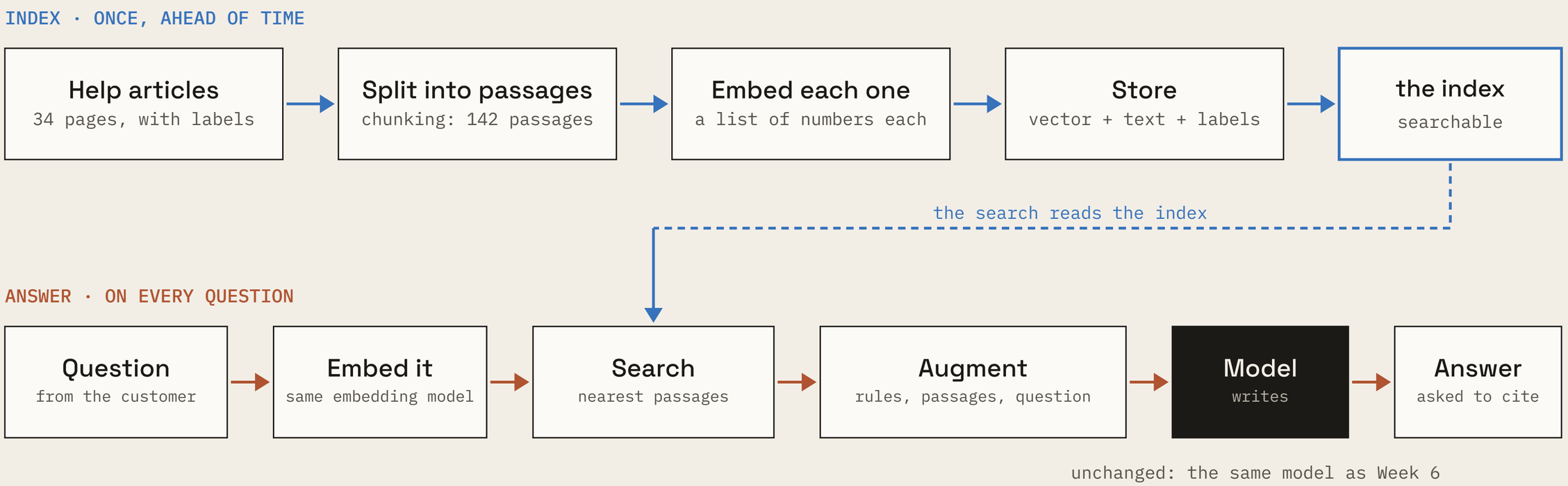
3 · IT DOES NOT SCALE

34 pages fit. Ten thousand?

About 306 tokens an article: 10,000 articles is 3.1 million tokens a call.

◆ 19 of the 27 answerable questions right, but slow, it gives out internal facts, and it cannot grow. **Choose a few pages, automatically.**

RAG: search first, then answer



- Two steps: **build an index once**, then **search it on every question** and put what you find in the prompt.

ONE ARTICLE, ON SHIPPING · 2,147 CHARACTERS · 21 SENTENCES

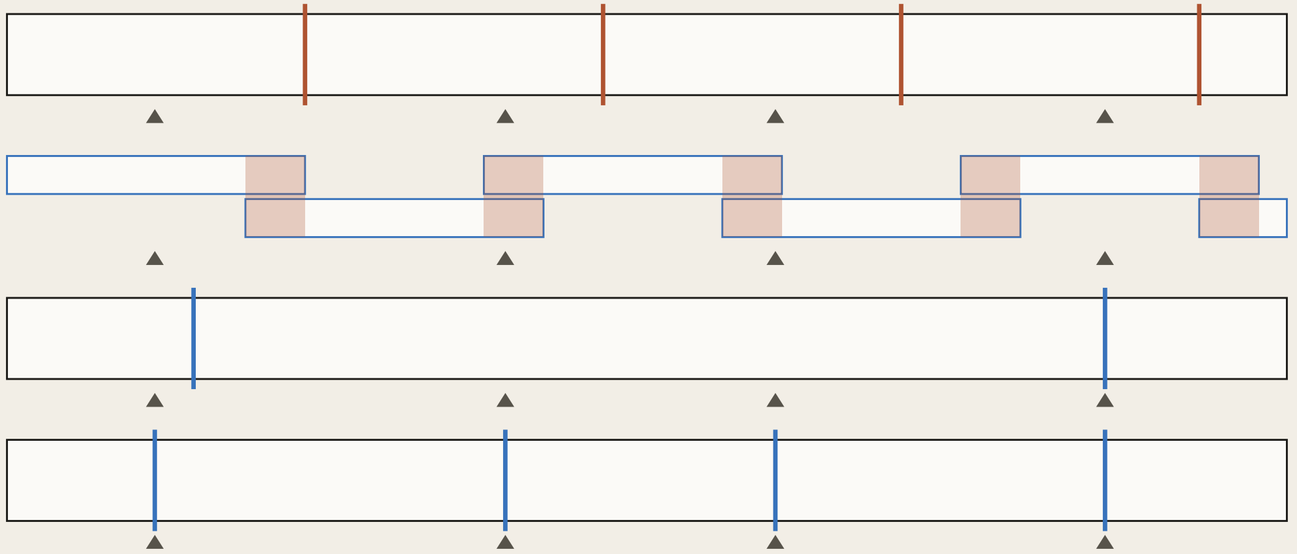
Chunking: four ways to split a page

FIXED SIZE
every 500 characters

FIXED SIZE, WITH OVERLAP
500, repeating the last 100

BY MEANING (SEMANTIC)
where the next sentence changes subject

BY STRUCTURE
at the writers' headings



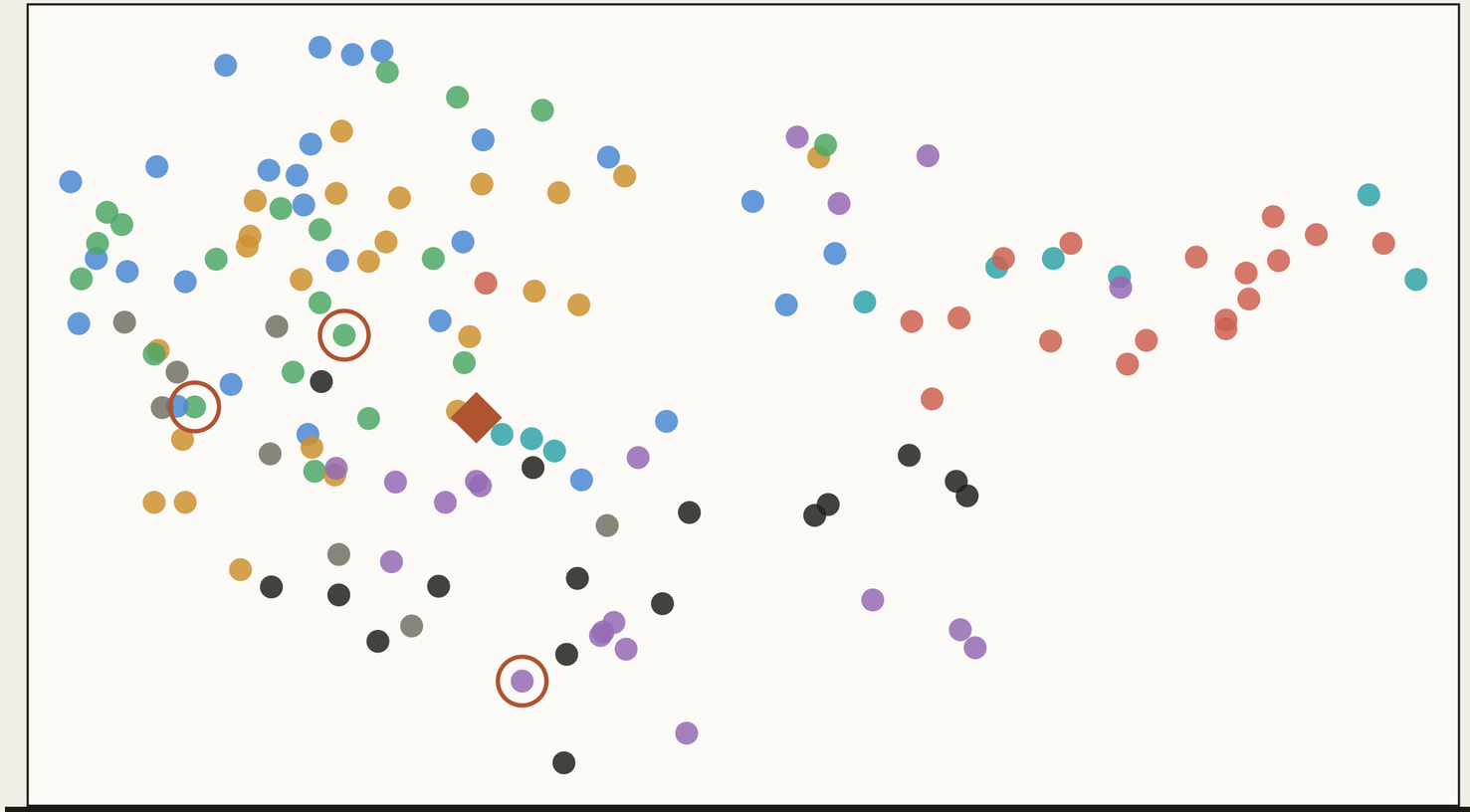
▲ a section heading in the article

THE FIRST FIXED-SIZE BOUNDARY: …ess day. Delivery windows below are counte ✂ d in business days from the day the parcel…

◆ A fixed-size split ignores the text and can cut a sentence in two. **Overlap repairs many of those cuts**. A semantic split follows changes of subject; a structure split follows the headings.

142 PASSAGES, 384 NUMBERS EACH WITH THIS MODEL, FLATTENED TO TWO

Embeddings turn text into numbers



- returns
- shipping
- billing
- account
- product
- warranty
- service
- internal
- the question
- its three nearest passages

THE QUESTION

The courier says it turned up but there is nothing on my doorstep. Now what?

HOW MANY NUMBERS? THE MODEL DECIDES

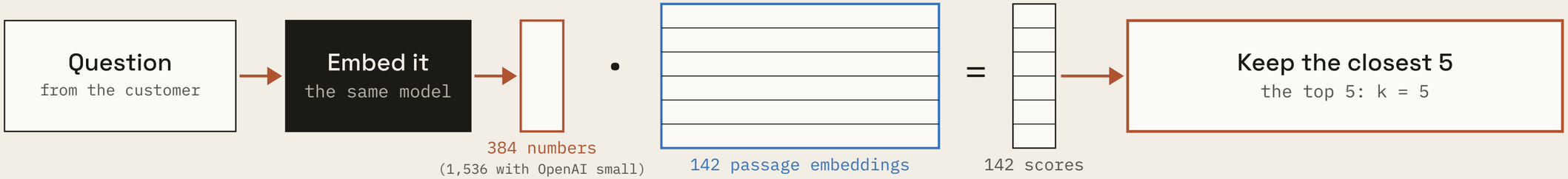
384 with the small local model used here; 1,536 with OpenAI's text-embedding-3-small. Use one model for questions and passages.

A SKETCH IN 2 DIMENSIONS (14% OF THE SPREAD): CLUSTERS ARE REAL, DISTANCES ARE NOT EXACT

- Texts that mean the same thing **land close together**, even when they share no words.

Search by meaning: find the closest passages

The question: ‘The courier says it turned up but there is nothing on my doorstep. Now what?’



SIMILARITY	ARTICLE · THE 5 CLOSEST PASSAGES	SECTION
0.525	Lost, Stolen, and Undelivered Packages	Marked delivered but not received
0.466	Lost, Stolen, and Undelivered Packages	Lost, Stolen, and Undelivered Packages
0.383	Suspicious Activity, Lockouts, and Two-Factor Sign-In	If you think someone else has been in your account
0.364	Tracking a Shipment	Reading a tracking status
0.334	Tracking a Shipment	When tracking has not moved

- Almost no words in common, yet the right page comes first: the search compared **meanings**.

Side note: what is an embedding model?

WHAT IT IS

Text in, one list of numbers out

Usually just the encoder part of Week 5's transformer. It has no decoder, so it produces a vector, not text: it cannot answer a question.

HOW IT IS TRAINED

On pairs of texts that belong together

A question and its answer, a title and its page, two wordings of one idea: hundreds of millions of pairs. Training pulls each pair's vectors together and pushes unrelated texts apart (contrastive learning).

MODEL	WHERE IT RUNS	VECTOR SIZE
all-MiniLM-L6-v2 (this lecture)	open · runs on a laptop	384
nomic-embed-text	open · runs in Ollama	768
BAAI bge-m3	open · laptop or GPU	1,024
OpenAI text-embedding-3-small	rented over an API	1,536
Cohere embed-v4	rented over an API	1,536
OpenAI text-embedding-3-large	rented over an API	3,072
Google gemini-embedding-001	rented over an API	3,072

DEFAULT SIZES, OCTOBER 2026 · OPENAI, COHERE AND GOOGLE CAN RETURN A SHORTER VECTOR

WHAT TO REMEMBER

Embed questions and passages with the same model. A new model means embedding every passage again. A longer vector is not automatically better.

- The embedding model decides the vector's length: **384 in this lecture, 1,536 with OpenAI's small model.**

Measure the search separately from the answers

RECALL@5 · THE RIGHT ARTICLE

Is a passage from the right article among the 5 returned?

Any section of that article counts. One score per question, then the average. **This search: 0.89**, so 89% of the articles needed were found.

FACT IN THE WINDOW · THE ANSWER ITSELF

Is the phrase that answers the question in those 5 passages?

The ‘window’ is what the model will read. Yes or no per question. **This search: 0.85**, so the answer was there for 23 of the 27 questions.

One question: ‘What will an Expedited order to Alaska cost me in shipping?’
Its answer is in two sections of the Shipping article: Fees and Surcharges.

THE FIVE PASSAGES THE SEARCH RETURNED

Shipping article **Fees section**	Shipping article **Surcharges section**	another article	Shipping article Addresses section	another article

RECALL@5: **YES** the Shipping article is among the five
FACT IN THE WINDOW: **YES** both sections with the answer are here

IF OTHER SECTIONS OF THE SAME ARTICLE HAD COME BACK

Shipping article Services section	another article	Shipping article Addresses section	another article	another article

RECALL@5: **YES** the Shipping article is still among the five
FACT IN THE WINDOW: **NO** the sections with the answer are missing

☐ from the right article: recall@5 counts it ▭ holds the answer: fact in the window needs it

- Recall@5: did the **right article** come back? Fact in the window: did **the answer** come back?

Which way of chunking works best?

SPLIT	PASSAGES	ANSWERS WHOLE (OF 27)	RECALL@5 (ARTICLE FOUND)	FACT IN WINDOW (ANSWER FOUND)	TOKENS IN WINDOW
200 characters	244	25	0.81	0.59	221
200 characters, 50 overlap	307	27	0.81	0.67	226
500 characters	111	26	0.91	0.81	489
500 characters, 100 overlap	120	27	0.91	0.89	525
by meaning (semantic)	113	27	0.91	0.81	576
one per section	**142**	**27**	**0.89**	**0.85**	**432**
one per article	34	27	0.91	0.93	1,543

THE SEMANTIC CUT HAS A SETTING

It embedded 458 sentences to choose its boundaries. Cutting at the lowest 20% / 33% / 45% of similarities gives fact in window 0.74 / 0.81 / 0.81.

THE EMBEDDING MODEL HAS A LIMIT

It reads the first 256 tokens of a passage and ignores the rest: 23 of 34 whole articles are longer, and 0 of 142 sections.

◆ Overlap repairs cut sentences. Splitting at the headings is **close to the best fixed-size split, for fewer tokens.**

THE SAME PASSAGES, RANKED TWO WAYS · RECALL@5

Two kinds of search: by keyword and by meaning

	EXACT CODES AND TERMS e.g. What does error code E-118 mean?	PARAPHRASES e.g. ‘nothing on my doorstep’
Keyword search ranks passages by shared words (BM25)	1.00 recall@5 **stronger**	0.12 recall@5 **much weaker**
Meaning search ranks passages by closest embedding	0.88 recall@5 **weaker**	1.00 recall@5 **stronger**

BM25 = ‘BEST MATCHING 25’, THE STANDARD KEYWORD SCORE

A passage scores for each question word it contains. Rare words count more, repeats count less each time, and a long passage gets no extra credit.

‘What is the Bulky Item badge on a product page?’

meaning search: the answering article is not in the top 5 · keyword search: found

‘The courier says it turned up but there is nothing on my doorste…’

keyword: not in the top 5 · meaning: rank 1

◆ Keyword search fails on paraphrases; meaning search is weaker on exact codes and terms. **Each is best where the other is worst: use both.**

Hybrid search: use both, then re-rank



SEARCH, TOP 5 PASSAGES	RECALL@5 (ARTICLE FOUND)	FACT IN WINDOW (ANSWER FOUND)
keyword search (BM25)	0.63	0.59
meaning search (embeddings)	0.89	0.85
both, merged by rank (RRF)	0.87	0.89
merged, then re-ranked	**0.91**	**0.93**

BM25: THE STANDARD KEYWORD SCORE

A passage scores for each question word it contains. Rare words count more, repeats count less each time, and a long passage gets no extra credit. (BM stands for ‘best matching’.)

RRF (RECIPROCAL RANK FUSION): HOW THE TWO LISTS ARE MERGED

Each passage earns points for its place in each list: the higher the place, the more points. The points from the two lists are added, and the passages are put in order of their totals. So a passage that both searches place near the top comes first: one placed 2nd and 3rd beats one placed 1st in one list and missing from the other. Places are used, not scores, because the two searches score in different units.

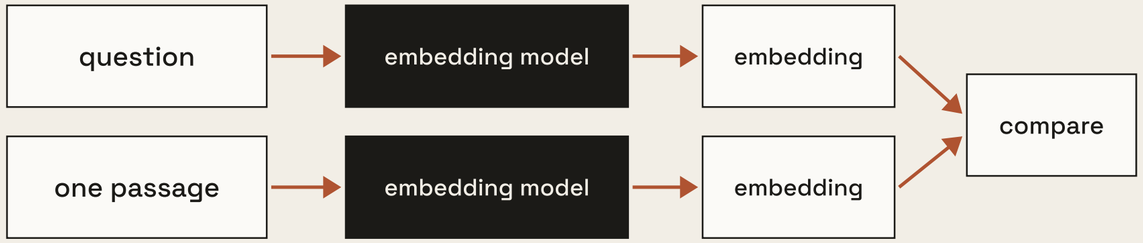
RE-RANKING

A second, slower model re-reads the 20 candidates with the question and keeps the best 5. The next slide explains it.

- Fact in the window: 0.85 with meaning alone → **0.93 fused and re-ranked.**

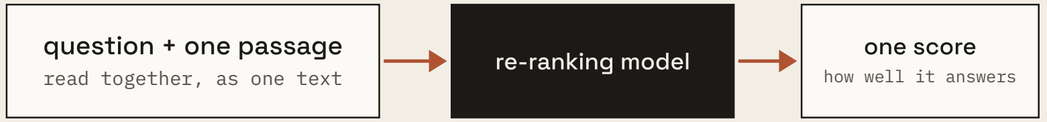
Re-ranking: a second model re-reads the top 20

STAGE 1 · THE SEARCH: FAST



read separately · passages embedded ahead of time

STAGE 2 · THE RE-RANKER: SLOWER, MORE ACCURATE



one run for each candidate · too slow to run on every passage

WHY NOT RE-RANK EVERY PASSAGE?

It would be too slow

It runs once for every passage it scores: fine for 20, not for all 142, or a million.

WHAT IT ADDED HERE

Fact in the window: 0.89 → 0.93

From the same 20 candidates, the answer reached the top 5 more often.

THE NAMES YOU WILL SEE

Bi-encoder and cross-encoder

Bi-encoder: the search's model, reads the two texts separately. Cross-encoder: a re-ranker, reads them together.

- A fast search finds 20 candidates; **the re-ranker reads each with the question** and keeps the best 5.

A vector database stores the embeddings

1 · KEEP

Vector, text and labels together

On disk, so nothing is embedded twice.

2 · SEARCH

The nearest vectors

With an index that checks only the most promising neighbourhoods: approximate by design, and what makes millions searchable.

3 · FILTER

By label, inside the search

audience = public AND status = current. Block 4 depends on it.

4 · CHANGE

One passage at a time

Add, update or delete an article without rebuilding the index.

NOTEBOOK §2.7 · OUR PASSAGES IN CHROMA · LAST: A QUESTION ABOUT AN INTERNAL CODE

```
stored: 142 passages, each with its vector, its text and three labels, in northwind_chroma/
same five passages in the same order as numpy: 27 of 27 questions
time per search: numpy 0.014 ms, Chroma 0.289 ms
no filter: the top 3 passages are all from an internal staff page
where audience=public AND status=current: the top 3 are all from public, current pages
a new Python process, opening the same folder, finds: 142 passages
```

OTHERS YOU WILL MEET · WHERE THEY RUN	EXAMPLES
inside your Python process	Chroma
a library: vectors and ids only	FAISS
inside your relational database	pgvector (PostgreSQL)
a server you run, or rent	Qdrant, Weaviate, Milvus
a service you rent	Pinecone

◆ At 142 passages NumPy and Chroma find the same five passages, and NumPy is quicker. **A vector database is worth having for storage on disk, filters and updates**, and at millions of passages for speed.

The augmented prompt: rules, sources, question

RULES · in the system message — **76 tokens**

only from the sources · cite the source id · say so if not there

SOURCES · in the user message · 5 passages — **509 tokens**

each labelled with its id: [KB-001] ×3 · [KB-006] · [KB-013]

QUESTION · in the user message, last — **13 tokens**

What will an Expedited order to Alaska cost me in shipping?

→ **Model**
unchanged

THE ANSWER, 0.9 S · RIGHT

An Expedited order to Alaska will cost you $12.95 plus a remote-destination surcharge of $14.95, totaling $27.90.

SYSTEM MESSAGE OR USER MESSAGE?

The rules are yours and the same on every call: they go in the system message. The sources and the question change on every call, and retrieved text is data, not instructions: they go in the user message.

- Rules first, then labelled sources, then the question. **Most of the prompt is sources**, the part the search controls.

Data problem 1: an out-of-date page

RETURN POLICY, VERSION 2 (OLD) — March 2023

30-day window
15% restocking fee
start a return by phone

audience: public · status: superseded

RETURN POLICY, VERSION 4 (CURRENT) — January 2026

60-day window
no restocking fee
start a return online

audience: public · status: current

To the search, both pages match the question equally well. Only the status label says which is current.

THE PROBLEM, MEASURED

old policy in 3 of 3 answers

With every page in the index, the old version matches a returns question as well as the new one.

THE FIX, MEASURED

old policy in 1 of 3 answers

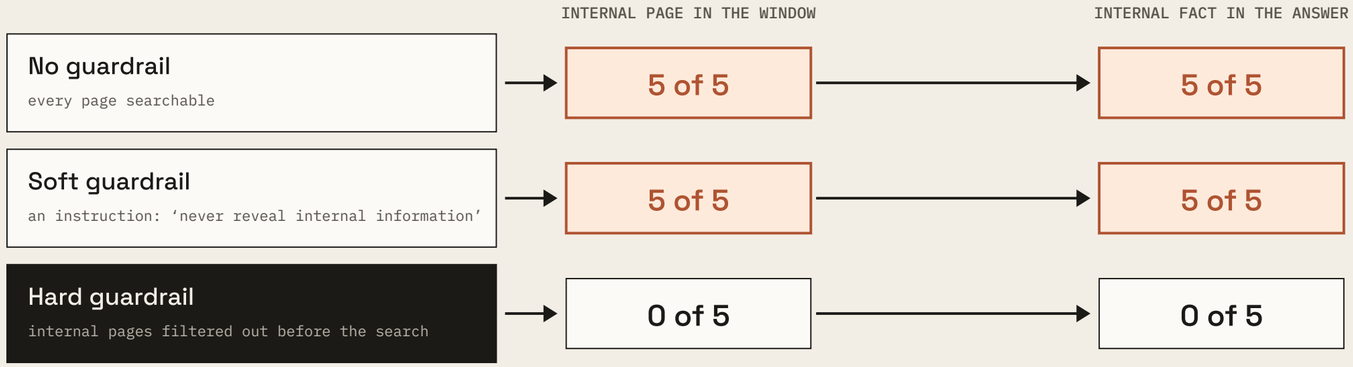
The search was limited to pages whose status is current. The one left came from a different cause: the search returned the wrong section.

WHAT TEAMS DO IN PRACTICE

Give every page a status and a date

Filter every search to current pages. When a page is replaced, take the old version out of the index the same day. Show the source and its date with each answer.

Data problem 2: internal pages



WHAT TEAMS DO IN PRACTICE

Check permissions in the search, not in the prompt

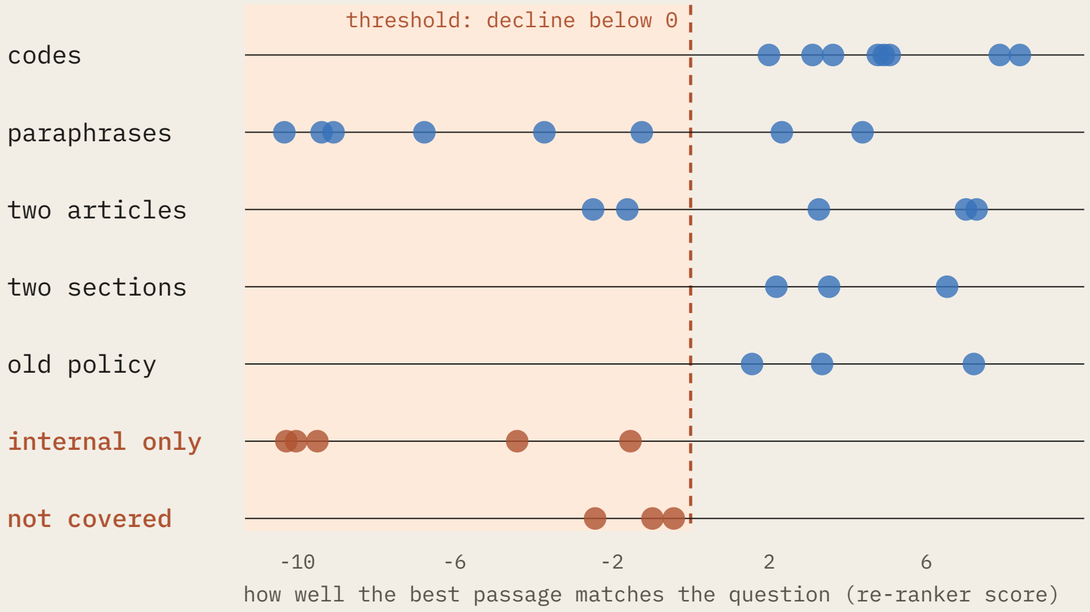
Label every page with who may see it. Filter each search by the role of the person asking: a customer’s search covers public pages only. Many teams also keep internal pages in a separate index.

◆ An instruction in the prompt did not stop the leak. **A page that is never retrieved cannot be leaked.**

8 OF THE 35 TEST QUESTIONS HAVE NO ANSWER THE ASSISTANT MAY GIVE

Data problem 3: no page has the answer

EACH DOT IS ONE TEST QUESTION, PLACED BY HOW WELL ITS BEST PASSAGE MATCHED



WHAT TEAMS DO IN PRACTICE

A threshold, then a hand-off

Before the model writes anything, look at how well the best passage matches the question. Below a threshold, do not answer: say so, and pass the customer to a person.

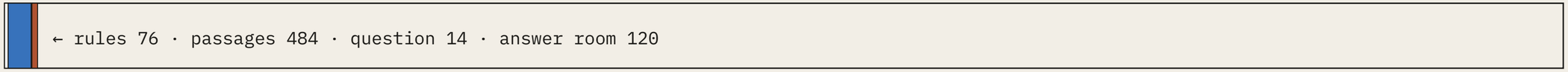
MEASURED ON THE 35 TEST QUESTIONS	DECLINED, OF THE 8 WITH NO ANSWER	TURNED AWAY, OF THE 27 REAL QUESTIONS
an instruction in the prompt only	4 of 8	0 of 27
plus: decline when the best match scores below 0	**8 of 8**	**8 of 27**

◆ A threshold stops made-up answers, and it also turns away some real questions. **Where to set it is a business decision.**

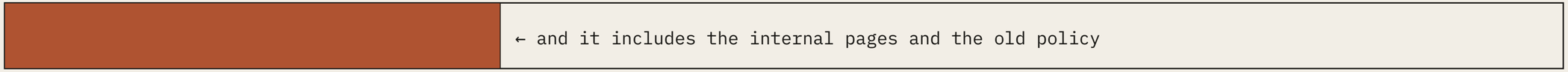
EVERYTHING THE MODEL READS ON ONE CALL, IN TOKENS

Context engineering: what goes into the prompt

ONE RAG CALL · 694 OF 32,768 TOKENS (2.1%)



PASTING THE WHOLE HELP CENTRE · 10,421 TOKENS OF SOURCES ON EVERY CALL (22× THE PASSAGES)



RULES

Stable, first

The same on every call.

PASSAGES

From the search

How many, and in what order: next.

CONVERSATION

Memory

What earlier turns to re-send.

FACTS

About this customer

Kept by the application, not the model.

ROOM

For the answer

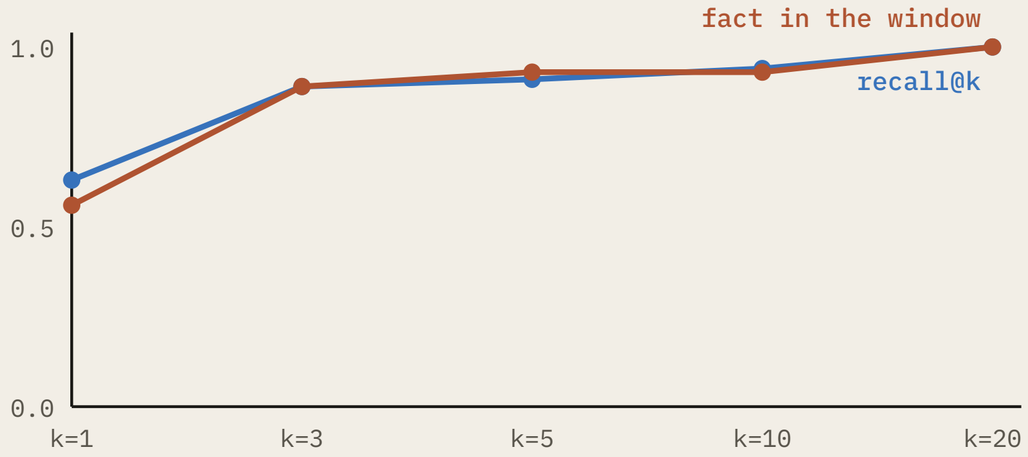
The reply needs space too.

◆ Context engineering: **choose what goes into the prompt** on every call. A large context window is a limit, not a goal.

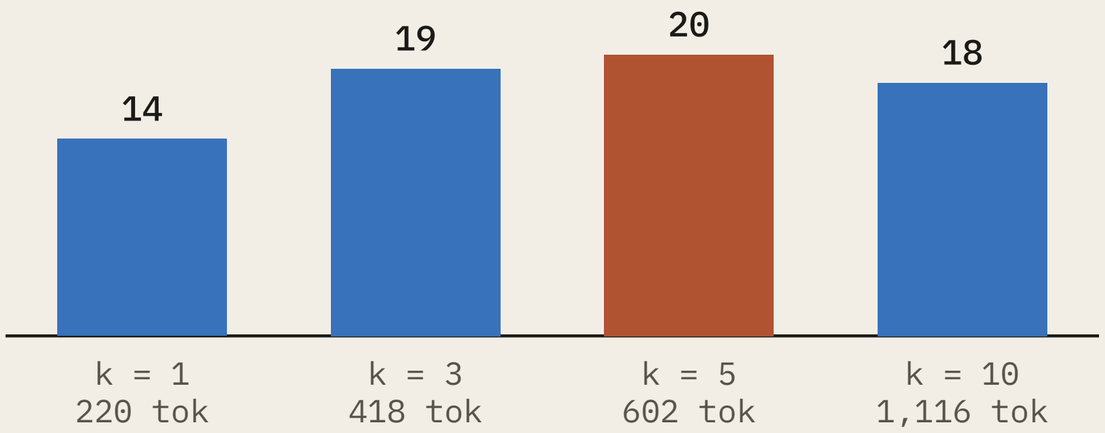
THE 27 ANSWERABLE QUESTIONS, WITH 1 TO 10 PASSAGES IN THE PROMPT

How many passages?

THE SEARCH (EVERY PAGE INDEXED) · RECALL@k AND FACT IN THE WINDOW



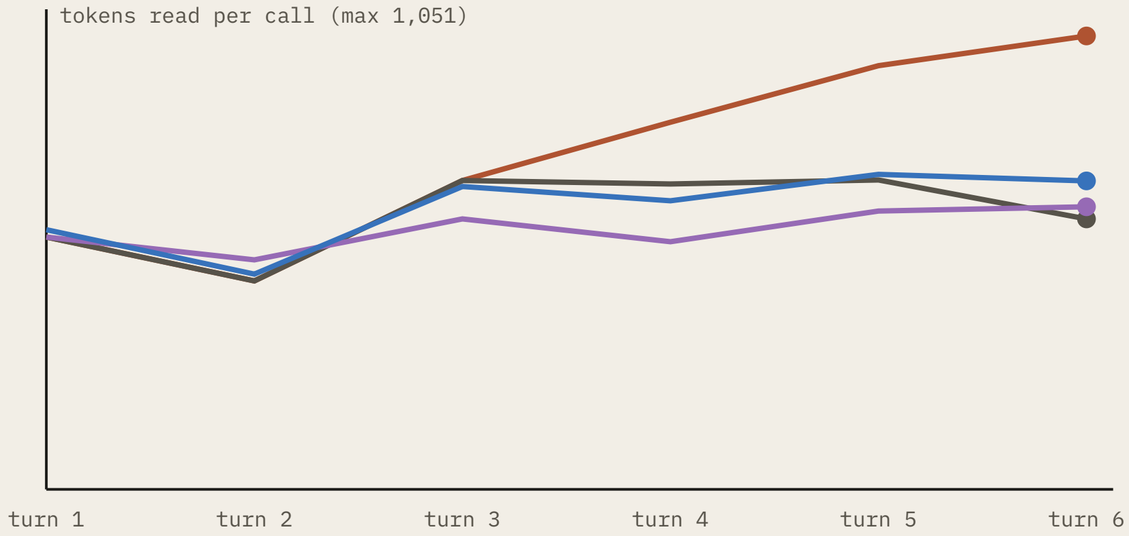
THE ANSWERS (PUBLIC, CURRENT PAGES) · RIGHT, OF 27



- Best at k = 5. **More passages did not mean better answers**, and every extra token is paid for.

SIX TURNS · AT TURN 6: ‘WHICH ORDER NUMBER WAS IT?’

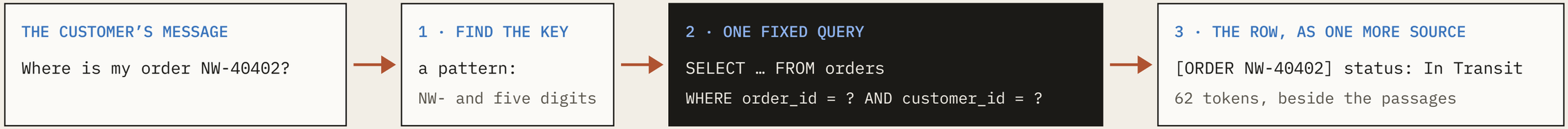
Conversation memory: what to re-send



keep everything	REMEMBERED	1,051 tok
last two turns	FORGOTTEN	627 tok
running summary	FORGOTTEN	655 tok
facts kept in code	REMEMBERED	715 tok

- Every earlier turn that is re-sent uses tokens. **Keep the key facts in the application’s code**; a summary can lose them.

Database lookup: facts that are in a table



■ the values are parameters, never part of the SQL ■ the connection is read-only ■ the query is scoped to the signed-in customer

5 THE TABLE CAN ANSWER, 2 IT MUST NOT	RIGHT	DECLINED
pages only	0 of 5	1 of 2
pages + the order's row	**5 of 5**	**2 of 2**

PAGES ONLY

Your order was shipped from Fulfillment Centre 1. To track this specific order, you should use the NWT-tracking …

WITH THE ROW

Your order NW-40402 is currently in transit. You received a tracking number via email shortly after placing the …

A ROW AND A RULE

'Can I still return order NW-39127?' It was delivered 104 days ago and the window is 60 days. The model said yes, with the rule in the window and without it; one line of SQL says no. **A comparison is the application's job.**

◆ A fact identified by a key, such as an order number, is **looked up, not searched for**: one fixed query, read-only, limited to the signed-in customer.

COST = TOKENS IN × INPUT PRICE + TOKENS OUT × OUTPUT PRICE

What each design costs, and how long it takes

US$ FOR 1,000 CUSTOMER QUESTIONS · 53 TOKENS OUT PER ANSWER · LIST PRICES CHECKED 2026-10-07

MODEL	LIST PRICE PER 1M TOKENS, IN / OUT	CLOSED BOOK · 58 TOKENS IN	RAG, TOP 5 · 568 TOKENS IN	PASTE EVERYTHING · 10,529 TOKENS IN
Claude Sonnet 5.5	$2.00 / $10.00	$0.65	$1.67	$21.59
GPT-6.1 sol	$2.00 / $10.00	$0.65	$1.67	$21.59
Gemini 3.1 Pro (preview)	$2.00 / $12.00	$0.75	$1.77	$21.69
Gemini 3.8 Flash	$0.75 / $3.75	$0.24	$0.62	$8.10
Claude Haiku 4.5 (older)	$1.00 / $5.00	$0.32	$0.83	$10.79
GPT-4.1 (older)	$2.00 / $8.00	$0.54	$1.56	$21.48

THE CALCULATION FOR ONE CELL

RAG on Claude Sonnet 5.5: 568 tokens in × 1,000 questions = 568,000 tokens × $2.00 per million = $1.14. 53 tokens out × 1,000 = 53,000 tokens × $10.00 per million = $0.53. Total $1.67 for 1,000 questions: about 0.17 cents a question.

WHERE THE PRICES COME FROM

claude.com/pricing · developers.openai.com/api/docs/pricing · ai.google.dev/gemini-api/docs/pricing. Token counts are the laptop model's; each provider counts tokens its own way, so these are estimates. On this laptop: 1.5, 1.1 and 14.1 s an answer.

- The design matters more than the model: pasting everything costs about **12–14× RAG**, on every question.

ALL 35 QUESTIONS: 27 TO ANSWER, 8 TO DECLINE

All the designs, compared

DESIGN	RIGHT	LEAKED	OLD POLICY	DECLINED	TOKENS / CALL	S / CALL
closed book	2/27	0/5	1/3	0/8	58	1.54
whole help centre pasted	19/27	5/5	1/3	1/8	10,529	14.07
RAG, every page indexed	18/27	5/5	3/3	1/8	568	1.15
RAG, filtered	**20/27**	**0/5**	**1/3**	**4/8**	**589**	**1.19**
RAG, filtered + answer checked	19/27	0/5	1/3	4/8	589	1.17

◆ No design is best in every column. Building a RAG system means **making a set of decisions** and measuring each one.

ASK WHAT IS MISSING: INSTRUCTIONS, FACTS, OR A BEHAVIOUR

Prompt, retrieval, or fine-tuning?

1 · CHANGE THE PROMPT

When you have not yet told the model clearly what you want

The task, the format, the tone, a few rules, a few examples. It costs minutes, so always try it first.

Example: ‘Sort this ticket into one of six queues and reply in this format.’

2 · ADD RETRIEVAL (RAG)

When the model is missing facts

Your company’s documents, policies and prices, and anything that changes. Search for the right passages and put them in the prompt; look records up in a database.

Example: ‘What is our return window?’

3 · FINE-TUNE THE MODEL

When it has the instructions and the facts and still does not behave as you need

A fixed format, style or classification, at high volume. It needs labelled examples and training, and retraining when the job changes. It is not a way to add facts.

Example: routing thousands of tickets a day into your own categories.

- Try them in this order: **the prompt first, retrieval when facts are missing, fine-tuning last.**

GROUP CASE · 7 MINUTES · CHOOSE, EVALUATE, CONTROL

Group case: the store-staff assistant

CHOOSE

One index, or two?

Staff may see the internal pages; customers may not. Filter by who is asking — where: in the prompt, or in the search?

EVALUATE

What would you test?

Which golden questions would you add for staff? What counts as a failure in each direction: a customer seeing an internal page, a staff member denied one?

CONTROL

What do you log and block?

What is recorded for each answer? What stops a staff member pasting an internal answer into a customer email?

- One assistant, two audiences: **what a person may see depends on who is asking**, and the filter is applied before anything reaches the prompt.

This week

ASSIGNMENT 7 · TEAM BUILD · DUE TUE OCT 13, 11:59 PM

Ship the help-centre assistant

A semantic cut you write, measured against sections · search and guardrails · memory and an order look-up in SQLite · ten test questions of your own · one improvement, measured · one final test on unseen questions · a chat page on Ollama. Built in a team GitHub repository: 15 of 100 points are for how you worked together in it, marked gently this first time. One submission per team, with a contribution statement.

PRACTICE · ON CANVAS

The same mechanics, a smaller help centre

The practice notebook and its solution, with a semantic cut and a table look-up to rehearse. Then the lecture notebook: every number on these slides is printed there.

FINAL PROJECT

Proposal due Tue Oct 13

Your project uses today's ideas: your own data, retrieved when a question arrives, with the search measured separately.

QUIZ 7 · WED OCT 14, START OF CLASS

This lecture and its notes

Six multiple choice, two fill-in, one written.

NEXT WEEK · WEEK 8

The model calls tools

Today the application ran the search and the order lookup on every question. Next week the model decides when to search and when to look something up.

Appendix: reference slides, not examined

A1

This lecture's split: one passage per section

How the help centre was split: at its headings.

A2

Reciprocal rank fusion: merging two ranked lists

Merging two ranked lists by rank, by hand.

A3

Checking an answer against its sources

Are the numbers in an answer also in its sources? Assignment 7 uses this check.

A4

Diagnosing a wrong answer: four questions

Find where a wrong answer came from.

A5

Does position in the prompt matter?

The answering passage first, in the middle, and last.

This lecture’s split: one passage per section

ONE ARTICLE, ON SHIPPING

Standard Shipping Options and Delivery Windows

Services and delivery windows

Fees

Surcharges

Addresses we cannot ship to

cut at each ##

5 PASSAGES · TOKENS EACH

# Standard Shipping Options and Delivery Windows	49
# Standard Shipping Options and Deli… ## Services and delivery windows	147
# Standard Shipping Options and Deli… ## Fees	116
# Standard Shipping Options and Deli… ## Surcharges	143
# Standard Shipping Options and Deli… ## Addresses we cannot ship to	77

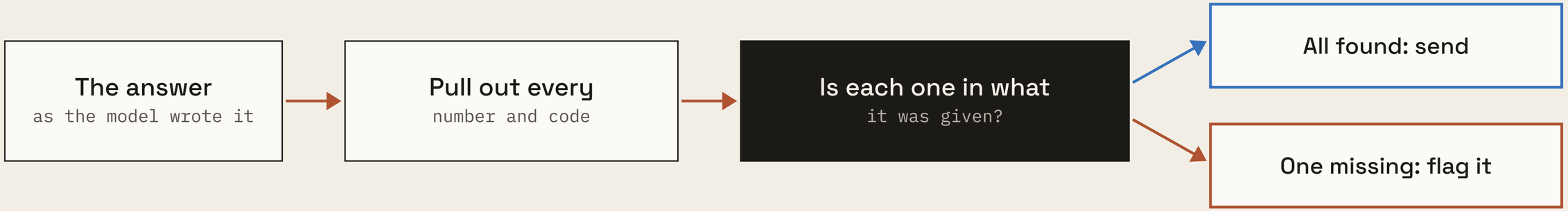
- Split at the headings: **one passage per section**, each with its article’s title in front. 142 passages, 22–177 tokens each.

Reciprocal rank fusion: merging two ranked lists

PASSAGE	KEYWORD RANK	MEANING RANK	FUSED SCORE
A	1	8	1/61 + 1/68 = 0.0311
B	**3**	**2**	**1/63 + 1/62 = 0.0320**
C	—	1	1/61 = 0.0164

```
def rrf(*rankings: list[int], c: int = 60) -> list[int]:
    """Reciprocal rank fusion: each list gives 1 / (c + rank); sort by the total."""
    total = defaultdict(float)
    for ranking in rankings:
        for rank, i in enumerate(ranking, start=1):
            total[i] += 1 / (c + rank)
    return [i for i, _ in sorted(total.items(), key=lambda kv: -kv[1])]
```

Checking an answer against its sources



TESTED ON 15 LABELLED ANSWERS

agrees on 15 of 15

Public, current pages indexed. With every page indexed: 14 of 15.

WHAT IT CANNOT CATCH

A leaked internal fact

If an internal page is in the prompt, every number in the leaked answer is on that page, so the check passes. Only keeping the page out of the prompt helps.

ON THE MODEL'S OWN ANSWERS

flagged 1 of 35

Its only flag: a correct total, $27.90, that the model added up itself, so it is on no page. A false alarm.

- Every number and code in the answer must be in its sources. **A free check**, but it cannot catch a leak.

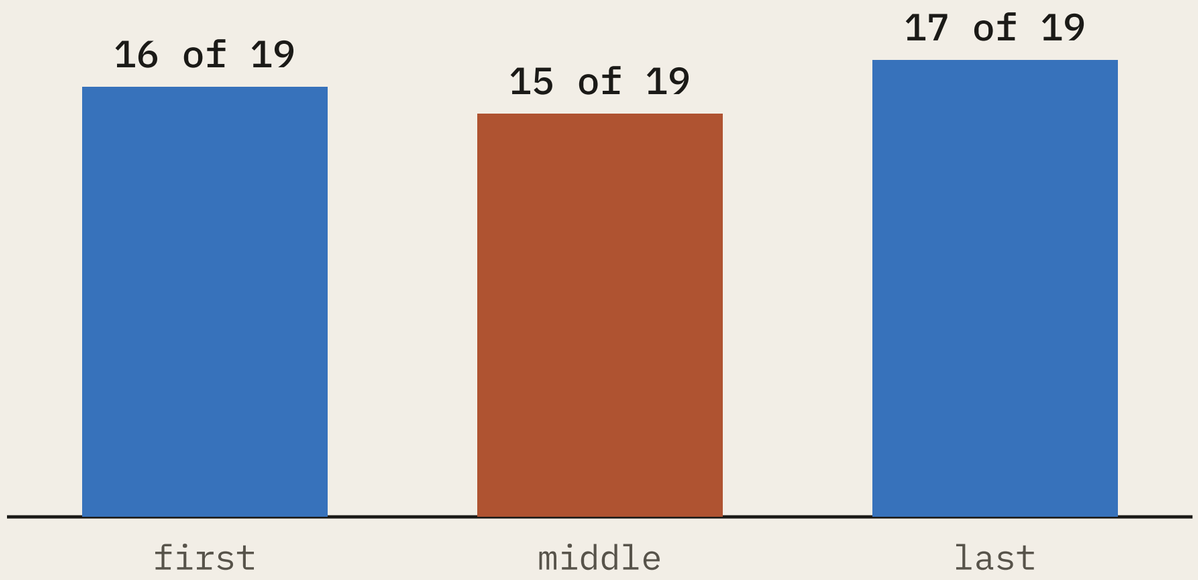
Diagnosing a wrong answer: four questions

1	**Was the fact in the window at all?** No: the search missed it	RETRIEVAL	fix: both searches, re-rank, more or fewer passages, better cuts
↓ yes			
2	**Is the answer on the pages it was given?** No: the model added something	MODEL	fix: cite and check (Block 3); clearer rules; less to read (Block 5)
↓ yes			
3	**Was the page it found the right page?** No: old, internal, or the wrong section	RETRIEVAL + DATA	fix: filters by status and audience, at search time (Block 4)
↓ yes			
4	**Does the right page exist, and is it current?** No: nobody wrote it, or nobody retired the old one	DATA FOUNDATION	fix: fix the documents: archive, label, write the missing article

◆ Check the prompt and the documents before you change the model. A search finds whatever is in the index, **including what should not be there**.

THE SAME TEN PASSAGES, THE ANSWER MOVED

Does position in the prompt matter?



THE TEST

19 questions; ten passages each, about 1,127 tokens; only the position of the answering passage changes.

THE RULE

Best evidence at the edges; keep the window short enough that there is little middle.

◆ Here position moved the score by one or two questions: a hint at laptop scale, a finding in long windows. **Order is a design decision.**